

PROJECT 1 SUMMARY: UK E-Commerce EDA

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Week: 1

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1. Project Overview

This project involved a comprehensive Exploratory Data Analysis (EDA) of a real-world UK-based e-commerce retail dataset. The dataset contained actual transactional data from a non-store online gift retailer spanning from December 2010 to December 2011. The objective was to transition raw, messy data into actionable business intelligence by cleaning the data, visualising trends, and answering critical business questions regarding revenue, product performance, customer value, and operational outliers.

2. Objective

The primary objectives of this task were to:

- Clean and prepare the dataset for analysis by handling missing values, duplicates, and invalid entries.
- Perform an in-depth EDA to uncover underlying patterns in customer behaviour and sales.
- Answer 8 structured business questions to diagnose the health of the business.
- Build a foundational narrative that would inform inventory management and marketing strategies.

3. My Approach

I adopted a structured 6-phase approach to ensure a logical flow from raw data to final insights:

- 1. Data Understanding:** Loaded the dataset and performed an initial sanity check to understand column types, missing values, and statistical distributions.
- 2. Data Cleaning (The "Make or Break" stage):**
 - a. Removed duplicate records.
 - b. Filtered out cancelled orders (negative quantities) to focus on actual sales.
 - c. Standardised date formats and text fields.

- d. Flagged missing Customer IDs to retain revenue data while isolating them for customer-specific analysis.
- 3. **Feature Engineering:** Created a new `'Revenue'` column (`'Quantity' × 'UnitPrice'`) and extracted time-based features (Month, Weekday, Hour) from the invoice date.
- 4. **Exploratory Analysis:** Visualised distributions, time-series trends (daily/monthly), and country-level performance using Matplotlib and Seaborn.
- 5. **Business Question Execution:** Utilised pandas aggregation functions to extract exact figures for top products, customers, and regions.
- 6. **Insight Synthesis:** Translated the numerical outputs into narrative business insights and strategic recommendations.

4. Tools Used

- **Programming Language:** Python 3.x
- **Environment:** Jupyter Notebook (Visual Studio Code)
- **Libraries:** Pandas (data manipulation), NumPy (numerical operations), Matplotlib & Seaborn (data visualisation), Datetime (date parsing).
- **Version Control:** Git (for local project tracking).
- **Visualisation Export:** Saved key charts as PNG images for integration into this report and the LinkedIn social post.

5. Challenges Encountered

- **Handling Missing Customer IDs:** Approximately 25% of the rows had missing `'CustomerID'` values. The challenge was retaining these rows for overall revenue calculations (as they represent actual sales) while excluding them from the "Top Customer" ranking to avoid skewing B2C metrics.
- **Inconsistent Negative Values:** The dataset contained negative quantities and prices, representing cancelled orders and administrative adjustments. Determining the correct threshold to filter these without losing valid wholesale bulk orders required careful analysis.
- **Memory Management:** The dataset (45 MB) was manageable, but performing complex group-bys on the Description column (4,000+ unique products) was a bit challenging and overwhelming, but it was doable.

6. Outcome (Key Insights)

- **Revenue Performance:** The total revenue generated during the period was £10.6 million, with a significant upward spike observed in November 2011, attributed to pre-Christmas buying.
- **Product Performance:** The top product by revenue was the DOTCOM POSTAGE with a record of £206.248.77, while a few high-volume, low-cost items (like PAPER CRAFT , LITTLE BIRDIE with a record of £ 80,995) dominated quantity sales.
- **Customer Concentration:** The top five customers by revenue collectively account for only ~11.8% of total revenue, with the single largest customer contributing just 3.15%. This indicates revenue is broadly distributed across the customer base rather than concentrated among a few dominant (wholesale/B2B) clients.
- **Geographic Insights:** While the UK dominates total sales, the Netherlands emerged as the highest-performing international market by average order value, presenting a clear expansion opportunity.
- **Data Quality Flags:** Identified over 200 "unusual" transactions with quantity > 100, confirming that these are legitimate B2B wholesale orders rather than errors, preventing inaccurate flagging by the operations team.

7. Lessons Learned

- **Context is Everything:** Treating high-quantity outliers as "errors" would have been a mistake. Understanding that the retailer sells to wholesalers allowed me to distinguish between bad data and valid B2B bulk purchases.
- **Documentation Saves Time:** Keeping a running log (Markdown cells in Jupyter) of every cleaning decision made the final report-writing phase significantly faster and more accurate.
- **Business Over Code:** I learned that a data analyst's job is not just to generate numbers (e.g., "Revenue is £X"), but to explain why that number matters to the CEO or Marketing Head.

8. Relevant Links

- **Jupyter Notebook (Source Code):** [Insert Google Drive Link to .ipynb or .html file]
- **EDA Visualizations:** [ EDA Visualisations]
- **LinkedIn Post:** [Insert Link to your mandatory social post]